



CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 20: Cylinder, Cone and Sphere (Surface Area and Volume)

Every formula and result of the chapter on one sheet

1	Cylinder: curved surface area	$2\pi rh$
2	Cylinder: total surface area	$2\pi r(r + h)$
3	Cylinder: volume	$\pi r^2 h$
4	Hollow cylinder: volume of material	$\pi(R^2 - r^2)h$
5	Hollow cylinder: total surface area	$2\pi(R + r)h + 2\pi(R^2 - r^2)$
6	Cone: slant height	$l = \sqrt{r^2 + h^2}$
7	Cone: curved surface area	πrl
8	Cone: total surface area	$\pi r(l + r)$
9	Cone: volume	$\frac{1}{3}\pi r^2 h$
10	Sphere: surface area	$4\pi r^2$
11	Sphere: volume	$\frac{4}{3}\pi r^3$
12	Hemisphere	Curved $2\pi r^2$; total $3\pi r^2$; volume $\frac{2}{3}\pi r^3$
13	Spherical shell: volume of material	$\frac{4}{3}\pi(R^3 - r^3)$
14	Recasting	Volume before = volume after
15	Units	1 litre = 1000 cm ³ ; 1 m ³ = 1000 litres